

Toy Blockchain Project Showcase

Rust-based blockchain prototype with consensus, networking, persistence, and API layers

What this project demonstrates

Why it matters

Core strengths

- Systems-level Rust implementation with clear ownership and concurrency boundaries.
- End-to-end blockchain logic covering blocks, UTXO state, Merkle roots, and mining.
- Networked architecture for peer sync, chain replacement, and transaction broadcasting.
- Persistent storage and API design that make the project practical to inspect and run.

Architecture highlights

- Consensus & block validation
- UTXO transaction model
- Merkle tree verification
- P2P synchronization logic
- REST API for node inspection

Notable implementation areas

- Rust modules for blockchain, network, mempool, and transactions
- Persistent storage using sled
- Cryptographic verification with ed25519 and SHA-256
- Test coverage for validation and chain behavior
- Seed data and example state for reproducible demos

Technology stack

■ Rust ■ Tokio ■ Axum ■ Serde ■ sled ■ SHA-256 ■ ed25519 ■ Merkle Trees ■ REST API ■ P2P

Why this should stand out

This project combines core software engineering skills—system design, data modeling, networking, cryptography, testing, and API development—into a single, runnable demonstration. It is especially strong for showing practical reasoning about reliability, correctness, and distributed behavior.